Using Evolutionary Game Theory to Understand Scalability in Task Allocation

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ABSTRACT

Cooperation is an important challenge in multi-agent systems. Decentralised learning of cooperation is difficult because interactions between agents make the environment non-stationary, and the reward structure tempts agents to act selfishly. A centralised solution bypasses these challenges, but may scale poorly with system size. Understanding this trade-off is important, but systematic comparisons have been limited to tasks with fully aligned incentives. We introduce a new task for studying cooperation: agents can solve the task by working together and specialising in different sub-tasks, or by working alone. Using neuroevolution, we empirically investigate scalability comparing centralised and decentralised approaches. A mathematical model based on the replicator dynamics allows us to further study how the task's social dynamics affect the emergent behaviour. Our results show that the task's unique social features, in particular the challenge of agents' physical coordination, causes both centralised and decentralised approaches to scale poorly. We conclude that mitigating this coordination challenge can improve scalability more than the choice of learning type.

CCS CONCEPTS

• **Computing methodologies** → **Multi-agent systems**; **Cooperation and coordination**; *Neural networks*; *Genetic algorithms*.

KEYWORDS

Multi-agent systems, Neuroevolution, Evolution strategies, Coevolution, Complex Systems

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1 INTRODUCTION

Cooperation is an increasingly important challenge in Multi-Agent Systems [5]. Autonomous agents using Reinforcement Learning (RL) or Neuroevolution [18] perform well when their interests are completely aligned or opposed [14]. Reward structures for cooperation are more challenging because self-interested autonomous agents are tempted to free-ride, resulting in convergence to undesirable equilibria without cooperation [1, 7]. We can solve this problem by using centralised incentives, but this may come at a cost, since decentralised multi-agent systems are more scalable [10].

In centralised multi-agent learning, a single policy controls all agents simultaneously. It can find the best solution for the team as a whole, but leads to exponentially growing complexity with team size. Decentralised learning, where agents learn unique policies concurrently, is expected to scale better with team size, but often presents convergence issues due to non-stationarity [13, 16]. This study aims to shed light on this trade-off.

Overall, the trade-offs between centralised and decentralised learning for cooperation are important but not well understood. Comparisons have been limited to applications where complex cooperation is not required. Previous studies overlook important elements of cooperation, like specialisation [10] and temptation to defect [10, 21], and recent works in related fields have found that decentralised systems do not have, in practice, many of the benefits that are intuitively expected [12, 20].

We do a systematic comparison of centralised and decentralised methods for learning cooperation. We use CMA-ES [11], because it aligns well with the characteristics of the problem [15] and has shown competitive results [19]. We choose our environment to highlight strategic aspects of cooperation (Section 2), so our findings should be valid for all learning methods. Finally, we use a theoretical model of the evolutionary process, based on the replicator dynamics.

2 THE SLOPE FORAGING PROBLEM

Cooperation tasks in the literature are often simple and lack some of the complex social elements of cooperation that systems are likely to encounter in the real world [e.g., 10, 17]. Other tasks may have social complexity but require agents to also learn high-level representations [2, 3], making it difficult to study cooperation.

For this research we implemented a task we call Slope Foraging. The literature provides multiple variations of this task [4, 6]. Our implementation is an extended version of the one introduced in [6]. It features some of the more complex social elements of cooperation, particularly specialisation and a risk of defection; but also has a relatively simple state and action space, allowing us to focus on the strategic challenges of cooperation. It is also an analogue for more sophisticated real-world robotics tasks, such as mining, search and rescue and warehouse management.

In Slope Foraging (Figure 1-Part B) resources spawn in a "source" area just beyond a slope, and agents spawn in a 'nest' at the bottom of the slope. Resources that agents drop on the slope slide into a "cache" area, where they remain until an agent collects them and brings them to the nest. The environment is discrete and 2D (Figure 1-Part A) to avoid the added layers of complexity introduced by

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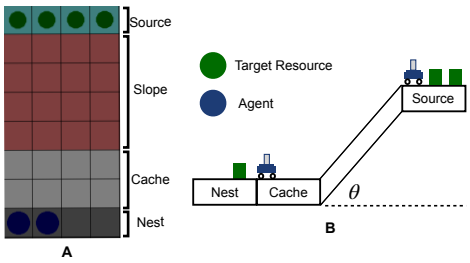

Figure 1: Slope Foraging arena for two agents

3D physics and continuous actions. To simulate the effects of a 3D slope, resources move in the slope area towards the cache at a faster rate than if they were carried by an agent. Resources can never slide into the nest. This simulates friction and prevents agents 'throwing' resources into the nest. Also, to simulate a robot battery, agents pay a higher cost travelling up the slope than moving elsewhere and pay a lower cost moving down the slope.

Agents can adopt a "Generalist" strategy, where they retrieve resources independently, or assume specialised "Dropper" and "Collector" roles, where a Dropper drops resources from the top of the slope and a Collector takes them from the cache to the nest. By cooperating, agents can retrieve more resources than they can alone and avoid the cost of repeated travel on the slope. From a game theory perspective, we can model this environment as a Stag Hunt game [8]. Cooperation earns agents a higher reward, but if an agent attempts to cooperate while its teammate acts selfishly, then it gets less reward than if it had acted selfishly as well. In addition to this challenge, pairs of agents that try to cooperate but do not coordinate their strategies (i.e. both are Droppers or both are Collectors) also get a worse reward than if they had been Generalists. We design the task to make cooperation beneficial but difficult to coordinate, introducing a temptation to defect and act selfishly.

Agents have a limited sensing range, to mimic real-world robots, making the problem more challenging. The supply of resources is infinite. In the decentralised setup, the reward for retrieving a resource is shared by all agents, regardless of their participation, but the costs of moving are borne individually. We describe the details of the environment's physics and cost function in the supplementary materials. We make our code available for others to use this environment[1].

3 REPLICATOR DYNAMICS

While experiments can provide empirical insights into how the different setups compare, implementation details can influence experimental results. A theoretical model of the learning process that abstracts away these details, such as solution representation or hyperparameters, allows us to focus on how Slope Foraging's social dynamics impact the observed results.

Evolutionary Game Theory (EGT) builds on classic Game Theory by introducing the concept of a population of self-replicating agents. The success, or fitness, of an agent's strategy depends on the distribution of strategies in the population. A successful agent replicates and the proportion of the population using its strategy increases [8]. Replicator dynamics are a tool from EGT that allows us to study an infinite population of replicating agents and observe what distribution of strategies is stable. We use the replicator equation as a model of learning.

In our model, agents can be Droppers, Collectors or Generalists, as in the experimental setup. We also include a strategy called Novice to capture the concept of agents that have not yet learned a policy. In a population of Generalists and Droppers, Dropper strategies will have lower than average fitness because there are no Collectors to pair with. Generalists will have higher fitness than average because they will succeed regardless of what strategy their team-mate uses. The proportion of Droppers will decrease and the proportion of Generalists will increase. We can observe whether the system converges to a population of generalists or cooperators.

When the teams are larger than two agents, the calculation becomes more complex. In a 4-agent version, the fitness of a Dropper is calculated by finding the probability of every 3-player combination of agents it can be grouped with, then calculating the average score across all combinations. We further explain the equation and how we apply it to Slope Foraging in the supplementary materials.

Given an initial distribution of strategies, a function to calculate agent fitness, a finite time horizon and the relevant model-specific parameters, the replicator equation calculates what distribution of strategies the population converges to. The final distribution, however, depends on the initial distribution, so it is useful to take the average over several initial distributions. By applying the replicator equation to a game, we can predict what equilibria a learning process converges to under specific assumptions.

4 RESULTS

4.1 Experimental Setup

We evaluate two learning types: centralised and decentralised. Centralised learning uses a single network to represent the joint-policy of a team. We represent a solution as a vector of the network's weights. Decentralised learning uses one network for each agent, with concurrent CMA-ES instances optimising each. Agents' best policies are stored in a shared data structure. Every generation, each agent's fitness is evaluated using a simulation, where the agent is partnered with the best representatives of all the other agents. Agents are evaluated separately to reduce the non-stationarity, as is standard for cooperative co-evolutionary algorithms [9].

We increase the team size and observe the rate at which fitness and specialisation degrade at scale. Neuroevolution is sample intensive so we limit team size to eight agents to ensure they are appropriately sampled, given our available computational resources. The trend of how the system scales is already visible for these team sizes (Section 4.2) and visual analysis of agent behaviours is easier than for very large teams. Further details of the network architecture and experimental setup are in the supplementary materials.

4.2 Scalability

As we expect, centralised learning has higher fitness than decentralised learning for teams of two agents (Figure 2). However, as team size increases, both setups scale similarly, instead of decentralised learning outperforming centralised learning. Both setups

[1]https://github.com/MostafaRizk/TS-Platform

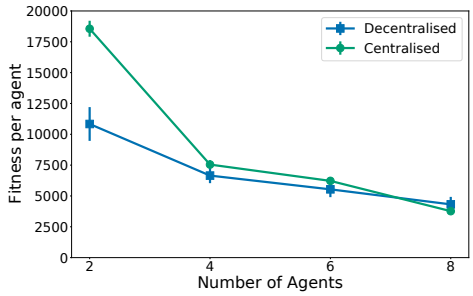

Figure 2: Scalability of fitness with number of agents

show a large drop in performance, between two- and four-agent teams, and steady decline for larger teams. This may be caused by the way the environment scales relative to agents' capabilities.

An individual agent has a limited sensing range, so different states can look the same from its perspective. The agent's recurrent network grants it a short term memory that helps it more accurately identify its current state based on its previous actions. However, the arena's width increases linearly with the team size. It is possible that when the arena is large, agents' memory is not sufficient to identify their state, making it difficult for Droppers and Collectors to align in the correct column. Although, as the team gets larger, agents are more likely to encounter partners, which makes cooperation easier. If these two factors scale at different rates, it may explain why the decline in fitness levels off. We investigate this further in Section 5.

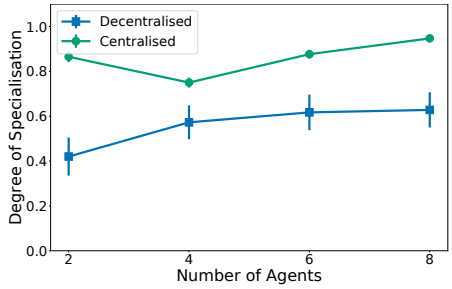

Figure 3: Scalability of specialisation with number of agents

Centralised learning finds teams with higher specialisation, as illustrated in Figure 3. We expect specialisation to decrease but, paradoxically, it increases for both setups. One possible explanation is that Slope Foraging allows for a form of 'partial cooperation' that does not require all agents to contribute. Two agents can form a Dropper-Collector pair while the rest are Generalists. As the teams get larger, there is more opportunity for a pair of agents to partner up and cooperate, but it is less likely for the whole team to cooperate. There is a good chance there will always be a subset of the team cooperating for teams of four or more, but as the teams get bigger, that fitness has to be shared with more agents, reducing the average fitness per agent. See video in the supplementary materials.

Another important element of Figures 2 and 3 is that teams evolved using the decentralised setup have a higher variance.

Investigating the evolved solutions, we observe three types of team behaviour: successful cooperators, successful generalists and 'failures of coordination' where agents attempt to cooperate but the Dropper and Collector are not aligned, so they retrieve no resources. Videos of all three behaviours are included in the supplementary materials. The centralised setup consistently learns specialisation, but solutions do not consistently perform well due to coordination failures. The decentralised setup experiences the same coordination challenges but additionally, a large number of runs converge to generalist solutions due to non-stationarity, so the variance is higher.

These findings suggest that the challenge of Droppers and Collectors aligning is inherent to the task. Even a centralised controller does not know where agents are in relation to one another and cannot always align them. However, decentralised learning has an additional challenge learning cooperation. At scale, these trends persist, but the decentralised setup learns cooperative solutions more often. This is why specialisation increases in Figure 3.

5 THEORETICAL MODEL

The experimental results show two unexpected trends: (i), fitness has a similar decline for all three learning types; and (ii), the degree of cooperation increases with team size even though the fitness decreases. Using the replicator dynamics (Section 3), we model the learning process to investigate the extent to which these findings result from Slope Foraging's social dynamics.

We define a payoff function to capture the game played by these different types of agent. This payoff also depends on the sliding speed of resources, i.e. the slope angle, and is designed to closely match the score that agents receive in the experimental setup. Details of how the payoff function is implemented are included in the supplementary materials.

We use the model to further investigate the two unexpected findings of our experimental results. To account for the decline in fitness for both setups, we hypothesise that limited sensing range poses an alignment challenge for the agents (see Section 4.2). To investigate this hypothesis, we include an alignment parameter p_a, which captures the raw probability of an agent pair aligning. This is the probability that a Dropper and Collector, in a team of two agents, will successfully align with each other in the same column. When $p_a = 1.0$, two agents always align successfully; when $p_a = 0$, they never align.

We investigate 5 different values for $p_a \in [0, 1]$. For each value, we sample 1000 different initial distributions and use the replicator equation (Section 3) to compute 1000 final distributions and calculate the average payoff per agent across those distributions. We repeat this across multiple team sizes to observe how the payoff changes at scale and predict that when $p_a < 1$, the payoff per agent will decrease with team size. Our results show that this is indeed the case. When $p_a = 1$, the payoff per agent scales constantly. If a Dropper and Collector are always able to align, every subsequent pair will increase the total payoff so that the payoff per agent remains the same. For $p_a = 0.9$ and $p_a = 0.5$, the payoff for a team of two agents is high but then rapidly declines for larger teams before levelling off. This is very similar to the trajectory of all three learning types observed in Figure 2. When the model accounts for the alignment challenge, it more accurately predicts the experimental results, supporting the hypothesis that the alignment challenge is

common to all three learning types and is the reason they scale similarly.

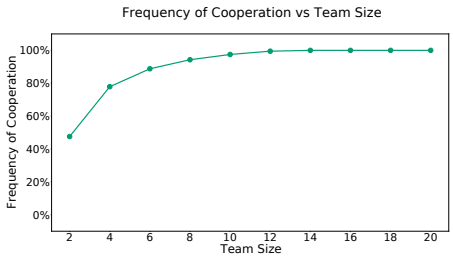

Figure 4: Frequency of convergence to each equilibrium using different team sizes and slope of size 8.

The second experimental finding we seek to investigate is the increase in cooperation despite the decline in fitness. If this is a result of the social dynamics, the model should predict a similar trend. We repeat the same process of sampling 1000 initial distributions and calculating the resulting equilibria. We then calculate the frequency with which the system converges to the cooperative equilibrium and repeat this for different team sizes. We use $p_a = 0.9$ as this achieves results closest to the experiments based on parameter tuning. Figure 4 shows that indeed, the model predicts that larger teams are more likely to converge to the cooperative equilibrium. This supports the hypothesis that the social dynamics of Slope Foraging's cause this trend. We suggested in Section 4.2 that the specific cause is that agents are able to partially cooperate. The increase in cooperation is most pronounced for the decentralised setup in Figure 3. Together, these findings suggest that decentralised learning may uniquely benefit from partial cooperation at scale.

6 DISCUSSION

Decentralised learning was expected to scale better with team size, but our results reveal a more nuanced reality. For Slope Foraging, all learning types show comparable declines in fitness at scale. Our experimental results, and model, show that the task's alignment challenge causes the decline. No learning type has a significant advantage at overcoming this challenge. Though we could not investigate larger teams, the theoretical model suggests that the alignment challenge will have less impact at larger scale. However, it lowers the scores of both setups significantly, so any advantage of one over the others is small. Addressing the alignment challenge is likely to have greater impact on scalability than the choice of learning type. Enhancing agent capabilities by implementing communication, theory of mind, or an internal model of the world, can address the challenge by allowing agents to locate one another more easily.

Another outcome of the study was the increase in cooperation at scale, which we found in both the experimental and theoretical model. Visual inspection of the evolved teams showed 'partial cooperation', where subsets of the team cooperate without the others. This phenomenon arises in Slope Foraging because each agent only requires one cooperator. We conclude that partial cooperation makes it easier for cooperation to emerge in large decentralised teams. Every cooperating agent has more opportunities to find a complementary cooperator, causing a snowball effect that leads to more evolutionary runs converging to cooperative solutions.

Overall, cooperation is complex. Our study shows that the subtleties in a task's social dynamics can dramatically impact the trade-offs between different learning types. We conclude that a task's social dynamics have a significant impact on the trade-offs of the different learning types and this is something future studies should investigate.

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